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4.1	$\langle \rangle - \overline{A_1}$	5
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1

$\langle\langle ()\rangle\rangle = \langle\beta, E, T, G, M\rangle, :$

- $\beta = : \langle\langle ()\rangle\rangle;$
- $E = ();$
- $T = -;$
- $G = ;$
- $M = , \quad E.$

$(\quad):$

$X = E = [0; 50]$

(1)

: $= 0 \quad (, \quad), \quad = 50 \quad (: , ,).$

2

- :

$T = \{ \langle\langle \rangle\rangle, \langle\langle \rangle\rangle, \langle\langle \rangle\rangle, \langle\langle \rangle\rangle \}$

(2)

$A_i \quad E = [0; 50] \quad \mu_{A_i}(x) \in [0; 1]. \quad (\sup \mu_{A_i}(x) = 1) \quad , \quad .$

1.

A_1	—	0	—	$[0; 5]$	7,5
A_2	—	5	10	15	20
A_3	—	15	20	25	30
A_4	—	25	30	$[35; 50]$	—

2.1 A_1 — «»

$()$:

$$\mu_{A_1}(x) = \begin{cases} 1, & x \leq 5, \\ \frac{10-x}{5}, & 5 < x \leq 10, \\ 0, & x > 10. \end{cases} \quad (3)$$

$: S_{A_1} = [0; 10)$. $: \text{kernel} A_1 = [0; 5]$. $: \text{pt} A_1 = \{7,5\}$.

2.2 A_2 — «»

:

$$\mu_{A_2}(x) = \begin{cases} 0, & x \leq 5, \\ \frac{x-5}{10}, & 5 < x \leq 15, \\ \frac{25-x}{10}, & 15 < x \leq 25, \\ 0, & x > 25. \end{cases} \quad (4)$$

$: S_{A_2} = (5; 25)$. $: \text{kernel} A_2 = \{15\}$. $: \text{pt} A_2 = \{10 ; 20\}$.

2.3 A_3 — «»

:

$$\mu_{A_3}(x) = \begin{cases} 0, & x \leq 15, \\ \frac{x-15}{10}, & 15 < x \leq 25, \\ \frac{35-x}{10}, & 25 < x \leq 35, \\ 0, & x > 35. \end{cases} \quad (5)$$

$: S_{A_3} = (15; 35)$. $: \text{kernel} A_3 = \{25\}$. $: \text{pt} A_3 = \{20 ; 30\}$.

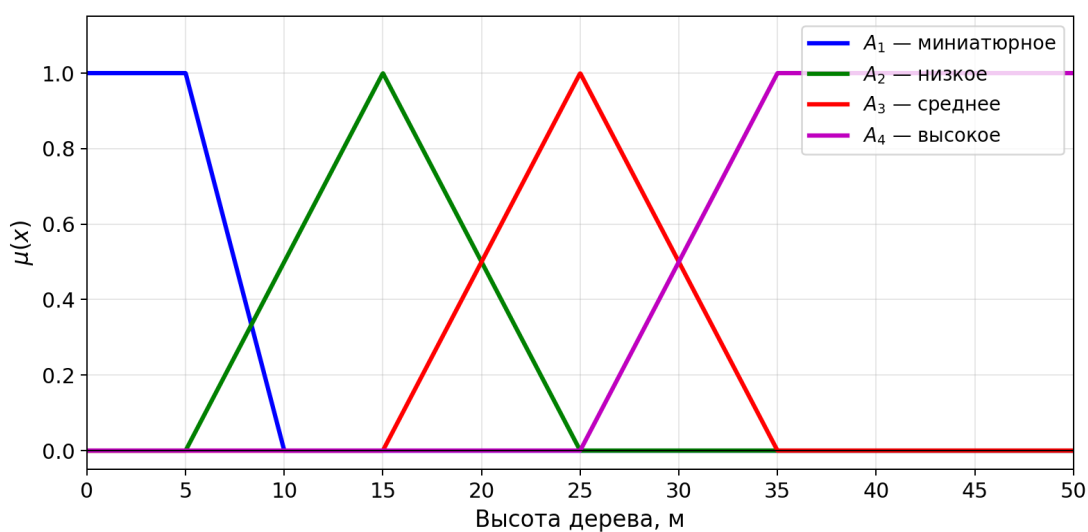
2.4 A_4 — «»

():

$$\mu_{A_4}(x) = \begin{cases} 0, & x \leq 25, \\ \frac{x-25}{10}, & 25 < x \leq 35, \\ 1, & x > 35. \end{cases} \quad (6)$$

: $S_{A_4} = (25; 50]$. : $\text{kernel} A_4 = [35; 50]$. : $\text{pt} A_4 = \{30\}$.

. 1.



. 1. « ()»: A_1 — «», A_2 — «», A_3 — «», A_4 — «»

3 «»

A_3 — «» .

3.1

A_3 — E , :

$$S_{A_3} = \text{supp} A_3 = \{x \mid \mu_{A_3}(x) > 0\} = (15; 35) \quad (7)$$

3.2

— $x \in E$, $\mu_{A_3}(x) = 0,5$:

- $(15 < x \leq 25)$:

$$\frac{x-15}{10} = 0,5 \Rightarrow x = 20$$

- $(25 < x \leq 35)$:

$$\frac{35-x}{10} = 0,5 \Rightarrow x = 30$$

$$\boxed{\text{pt}A_3 = \{20 ; 30\}} \quad (8)$$

3.3

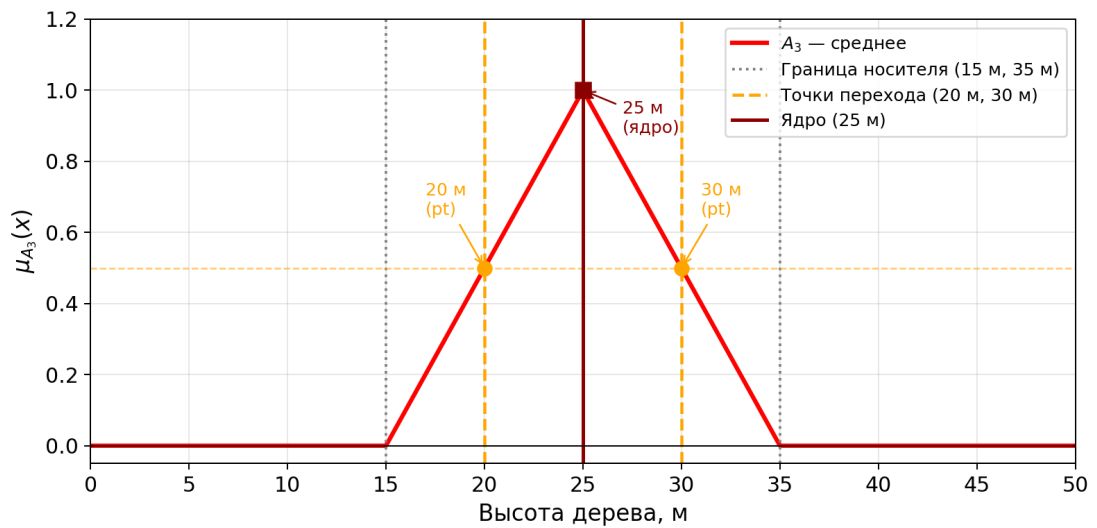
$$\text{kernel}A_3 = \{x \mid \mu_{A_3}(x) = 1\} = \{25\} \quad (9)$$

: 1 $x = 25$.

3.4

2. A_3 — «»	
S_{A_3}	(15 ; 35)
$\text{pt}A_3$	{20 ; 30 }
$\text{kernel}A_3$	{25 }
h	1 ()

«» . 2.



. 2. A_3 — «»: $(15; 35)$, $\{20; 30\}$, $\{25\}$

4

$G.$ M :

- «» — : $\mu_{\bar{A}}(x) = 1 - \mu_A(x)$;
- «» — : $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$;
- «» — : $\mu_{\text{CON}(A)}(x) = [\mu_A(x)]^2$.

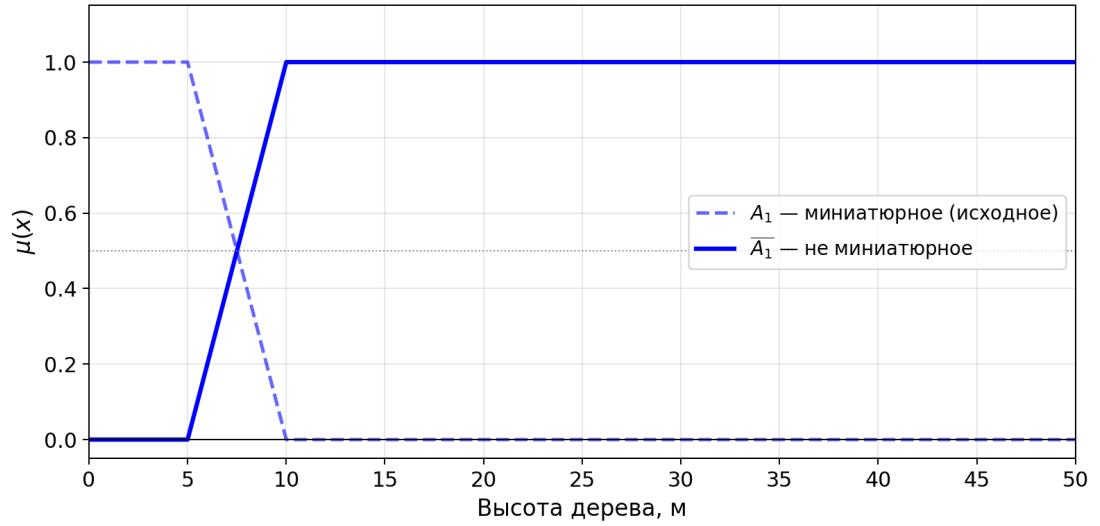
4.1 « » — $\bar{A}_1$

: .

$$\mu_{\bar{A}_1}(x) = 1 - \mu_{A_1}(x) = \begin{cases} 0, & x \leq 5, \\ \frac{x-5}{5}, & 5 < x \leq 10, \\ 1, & x > 10. \end{cases} \quad (10)$$

: 1. 10 $(\mu = 1)$.

. 3.



. 3. « » ($\overline{A_1}$): — $\overline{A_1}$, — A_1

4.2 « » — $A_1 \cup \overline{A_2}$

: , .

1. $\overline{A_2}$ (« »):

$$\mu_{\overline{A_2}}(x) = 1 - \mu_{A_2}(x) = \begin{cases} 1, & x \leq 5, \\ \frac{15-x}{10}, & 5 < x \leq 15, \\ \frac{x-15}{10}, & 15 < x \leq 25, \\ 1, & x > 25. \end{cases} \quad (11)$$

2. A_1 :

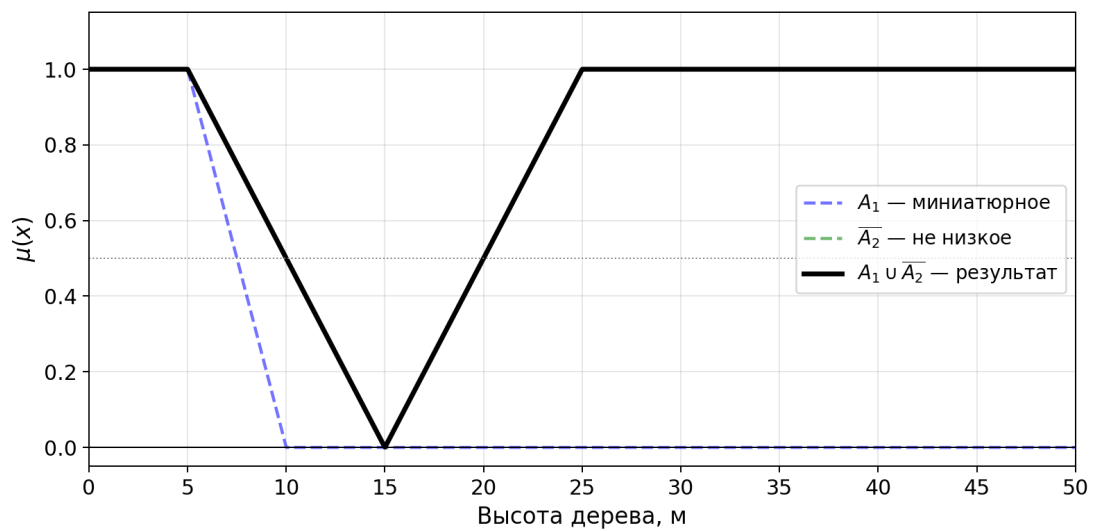
$$\mu_{A_1 \cup \overline{A_2}}(x) = \max(\mu_{A_1}(x), \mu_{\overline{A_2}}(x)) \quad (12)$$

3. $\mu_{A_1 \cup \overline{A_2}}(x)$

x	$\mu_{A_1}(x)$	$\mu_{\overline{A_2}}(x)$	$\max(\cdot)$
$x \leq 5$	1	1	1
$5 < x \leq 10$	$\frac{10-x}{5}$	$\frac{15-x}{10}$	$\frac{10-x}{5}$
$10 < x \leq 15$	0	$\frac{15-x}{10}$	$\frac{15-x}{10}$
$15 < x \leq 25$	0	$\frac{x-15}{10}$	$\frac{x-15}{10}$
$x > 25$	0	1	1

: $(x \leq 5 \vee x > 25) \rightarrow 5-25$.

. 4.



. 4. « » $(A_1 \cup \overline{A_2})$: —, — A_1 $\overline{A_2}$

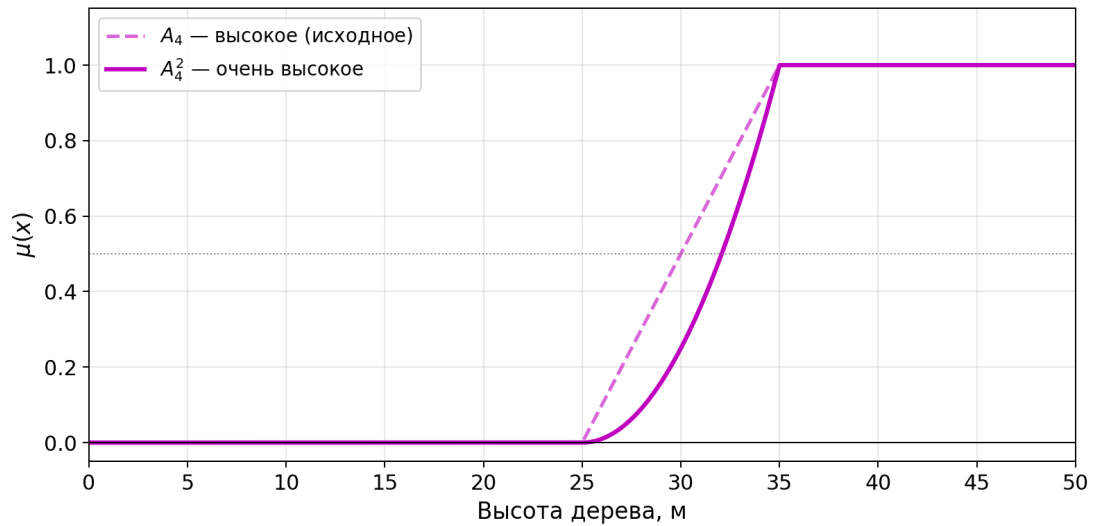
4.3 « » — $\text{CON}(A_4) = A_4^2$

: $(\alpha = 2)$.

$$\mu_{\text{CON}(A_4)}(x) = [\mu_{A_4}(x)]^2 = \begin{cases} 0, & x \leq 25, \\ \left(\frac{x-25}{10}\right)^2, & 25 < x \leq 35, \\ 1, & x > 35. \end{cases} \quad (13)$$

: $() \rightarrow A_4 [25; 35]$.

. 5.



. 5. « » ($\text{CON}(A_4) = A_4^2$): — A_4^2 , — A_4

4.4 « » — $A_2 \cup A_3$

: .

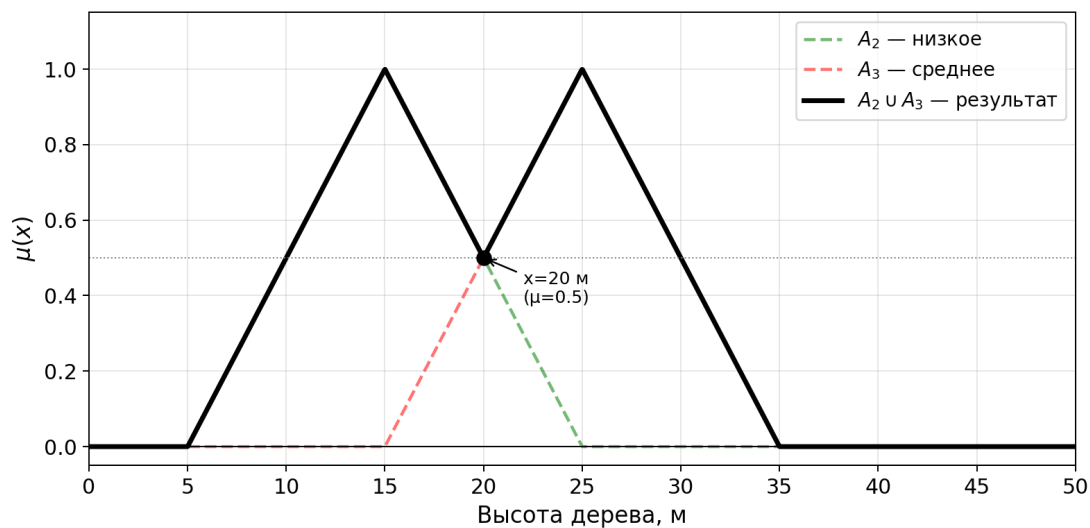
$$\mu_{A_2 \cup A_3}(x) = \max(\mu_{A_2}(x), \mu_{A_3}(x)) \quad (14)$$

4. $\mu_{A_2 \cup A_3}(x)$

x	$\mu_{A_2}(x)$	$\mu_{A_3}(x)$	$\max(\cdot)$
$x \leq 5$	0	0	0
$5 < x \leq 15$	$\frac{x-5}{10}$	0	$\frac{x-5}{10}$
$15 < x \leq 20$	$\frac{25-x}{10}$	$\frac{x-15}{10}$	$\frac{25-x}{10}$
$x = 20$	0,5	0,5	0,5
$20 < x \leq 25$	$\frac{25-x}{10}$	$\frac{x-15}{10}$	$\frac{x-15}{10}$
$25 < x \leq 35$	0	$\frac{35-x}{10}$	$\frac{35-x}{10}$
$x > 35$	0	0	0

: 5 35 $\mu = 1$ $x = 15$ $x = 25$, $\mu = 0,5$ $x = 20$.

. 6.



6. « » $(A_2 \cup A_3)$: — , — A_2 A_3

5

5.

(A_1)	—	, $[0; 10)$
(A_2)	—	, $(5; 25)$
(A_3)	—	, $(15; 35)$
(A_4)	—	, $(25; 50]$

	$\mu = 1 - \mu_{A_1}(x)$
+	$\mu = \max(\mu_{A_1}(x), 1 - \mu_{A_2}(x))$
$(\alpha = 2)$	$\mu = [\mu_{A_4}(x)]^2$
	$\mu = \max(\mu_{A_2}(x), \mu_{A_3}(x))$
